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Faculty of Engineering
Department of Electronics & Telecommunication
Engineering

EN1190: Engineering Design Project

Team – Titan Force

Smart Elephant Fence Breach Detector



Group Members

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1.Problem Description

1.1 Problem

Human-elephant conflict is a significant problem in Sri Lanka. It results in a substantial loss of human lives (an estimated 70–100 lives per year, according to IUCN/World Wildlife Fund), elephant lives (an estimated 20–25 per year), and causes major property damage to farmers and villagers (estimated at over 15 million LKR annually).

To address this issue, the Sri Lankan government allocates a considerable amount of money and resources. Existing methods to manage the conflict include electric fences, watch huts, trenches, noise and light deterrents, etc. Among these, **electric fences play the most prominent role** in preventing conflict.

Most of the areas affected by this conflict are surrounded by electric fences. However, elephants still manage to breach these fences and enter villages and farmlands. These incidents frequently occur at night, making it difficult for villagers and farmers to detect the intrusion until after the damage is done. **Our team has addressed this issue with a simple electronic device.**

1.2 Expected Goal

Our goal is to alert authorities, farmers, and villagers as soon as a fence is breached. We prioritize alerting the authorities first, as they are the key resource personnel capable of taking appropriate action to protect both humans and elephants when such breaches occur.

1.3 Solution

Our solution is an electronic device that will be installed on electric fence poles (one device for every 10 poles where there is an elephant path crossing the border or village, or predetermined places where elephants spend most of their time). To alert authorities through a message, the device makes a standard voice call, while villagers and farmers are notified through our custom mobile app.

2. Product Feasibility

2.1 Technical Feasibility

We designed our product to withstand the harsh environmental conditions in those areas, as well as the physical impact caused by elephant attacks. To keep the product cost-effective, we used affordable yet reliable sensors, a GSM SIM module for connectivity (since 2G is sufficient for its operation), and an ESP32 chip for processing tasks. The device is powered by reliable batteries that can last up to 24 hours and are recharged using solar energy.

2.2 Hardware Feasibility

The hardware components we used are,

- ESP 32D chip
- Solar Panels
- TP 5100 Battery Charger Module
- MP 1584 Voltage Converter Modules
- AP 2112 Linear Voltage Regulator
- BMI160 Accelerometer and Gyroscope Module
- SIM 900A Module
- 3.4V 1800mAh Li-ion Battery

2.3 Software & Online Platform Feasibility

- Arduino – To program the ESP-32 chip
- Android Studio – To create the mobile app
- Firebase – To create an online database and support cloud messaging
- Oracle – Server

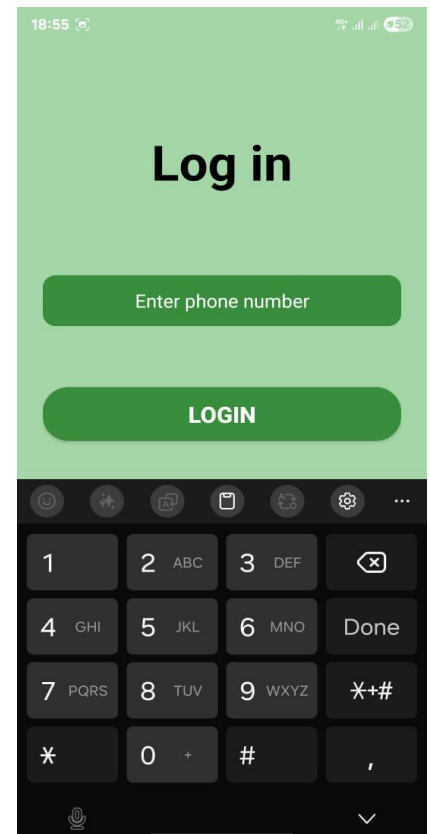
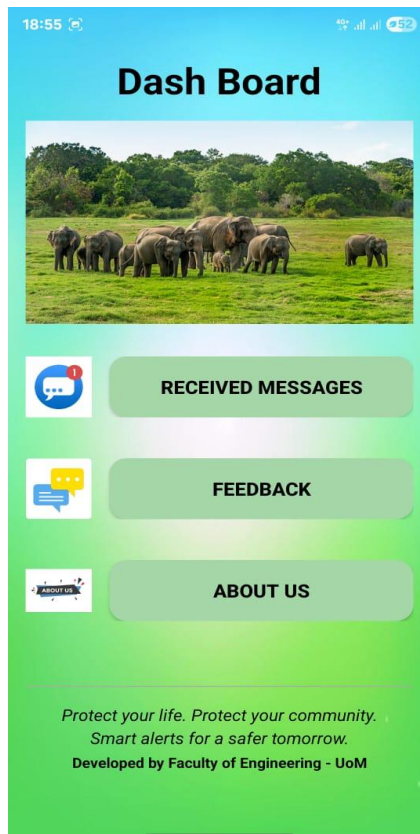
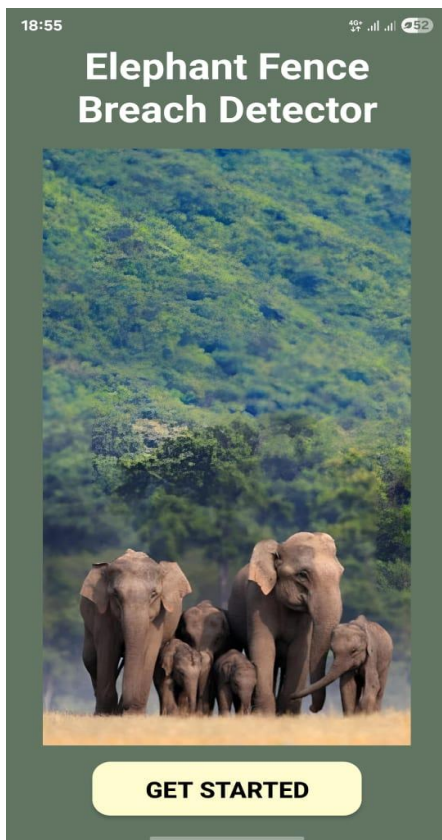
3. Product Applications

Our device will be installed on the fence poles. When it tilts from the initial position, rather than a threshold gyroscope module will detect that and an alert will happen. Since alerting authorities is more critical, we use standard voice calls that will be made directly through the GSM SIM module. Parallel to that, we alert farmers and villagers through a mobile app.

3.1 Mobile App

The Elephant Fence Breach Detector has several Features.

- Location can be detected
- Real-Time Notifications
- Movements can be detected
- Can store the data for future use



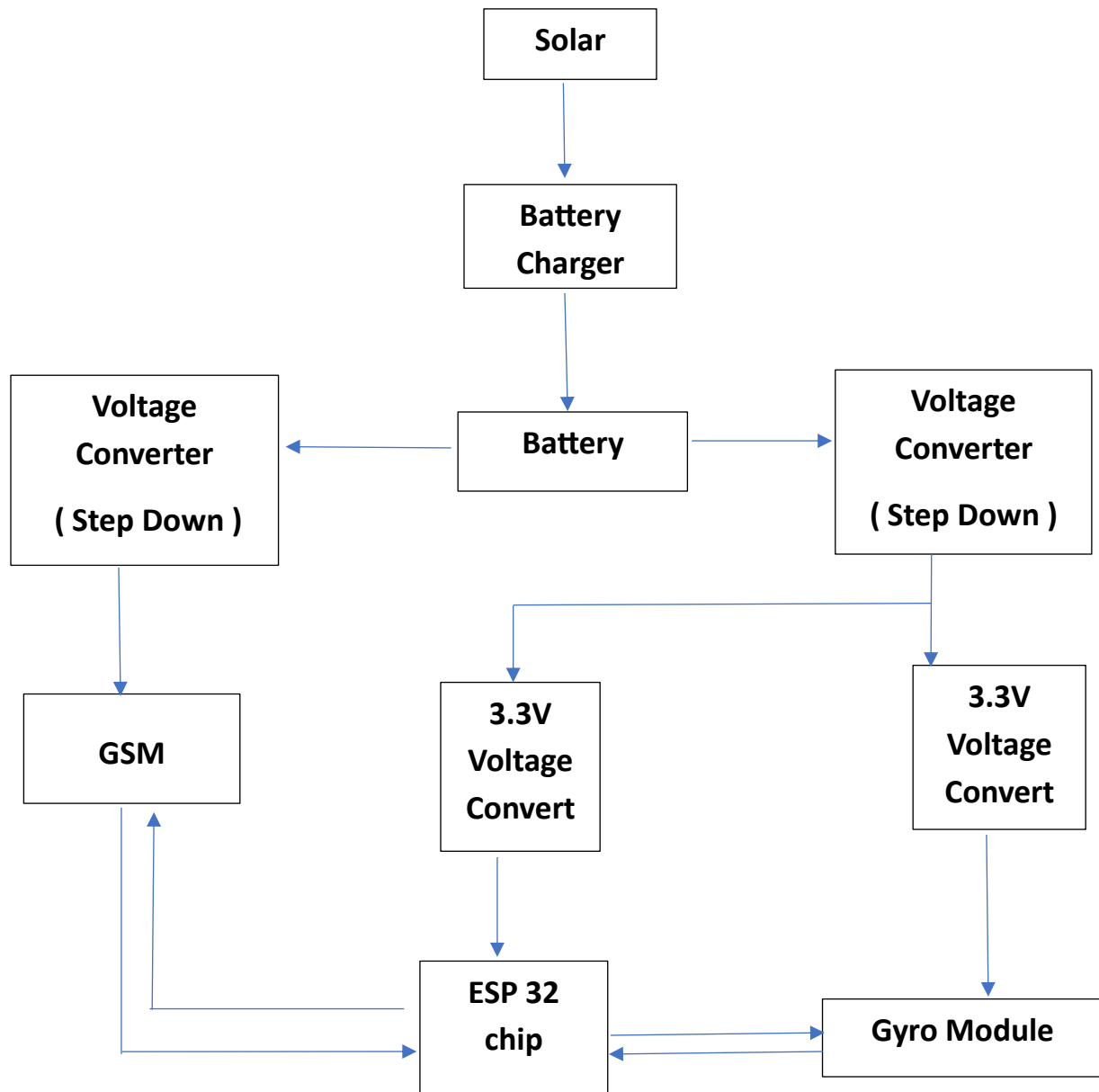
4. Product Specifications

- **Dimensions** : 80x60x130 cm
- **Weight**: 350g
- **Connectivity** – Supports 2G / 3G / 4G networks
- **Power Source** – Solar-powered for uninterrupted operation
- **Durability** – Waterproof and shock-resistant design
- **Installation** – Easy to install and configure
- **User Experience** – Reliable, clear, and user-friendly interface

5. Product Architecture

- **Power Supply** – The device is powered through two 3.4V 1800mAh rechargeable batteries, a TP 5100 battery charging module for charging the batteries and protecting them from overcharging.
- **Solar Power** - Two 5V, 80mA Solar Panels were used to charge the rechargeable Batteries.
- **Voltage Regulation** – Two MP1584 voltage converters are used to step down the voltage to 7.4V to 5V. One is used to supply 5V to power the ESP32D chip, and the other is used to supply 5V to the BMI160 gyro-accelerometer module independently.
- **ESP 32D Chip** – This Chip is used to check whether the conditions were true, and if true and is detected, the chip is responsible for sending an HTTP request and SMS via the SIM module
- **AP2112 Module** – Module is a low-dropout (LDO) linear voltage regulator designed for powering low-noise, low-current digital circuits. That is used in PCB to convert to 5v in 3.3 to power the ESP chip
- **BMI160** - The BMI160 is a low-power, high-performance inertial measurement unit (IMU) that combines a 3-axis accelerometer and a 3-axis gyroscope in a single package used to calculate the Rolling and Pitching angle of the Fence Pole.
- **SIM900A** – This Module is a dual-band GSM/GPRS module used to send SMS, make voice calls, and send an HTTP request to the Server

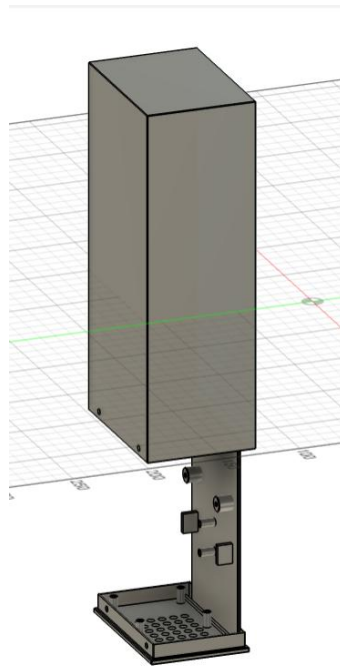
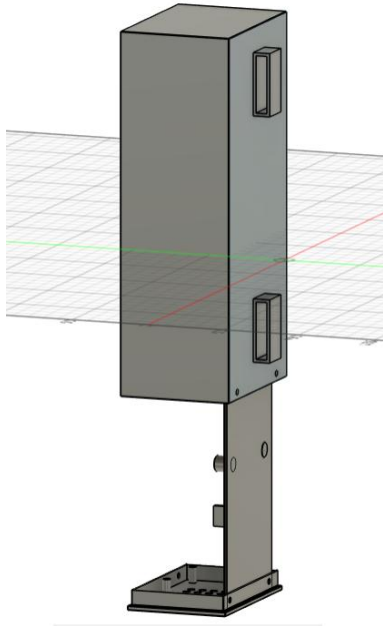
5.1 Block Diagram



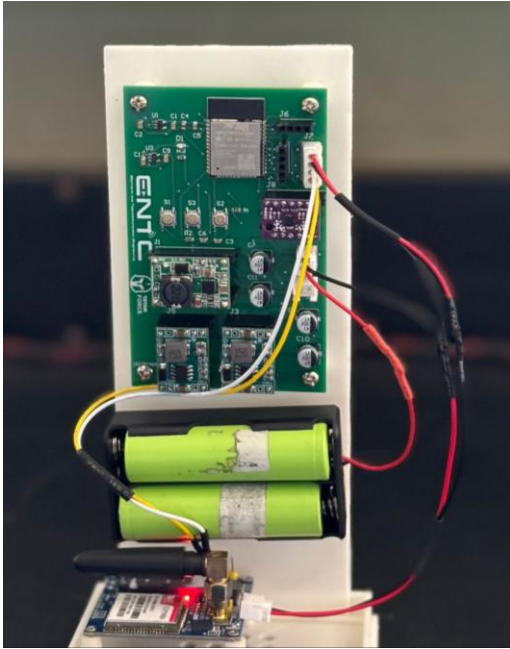
6. Product Enclosure Design

Since this device will be installed outdoors and may be subjected to physical shocks, we focused on a design that can withstand both environmental and mechanical stress. For the initial concept, we were inspired by the design of CCTV camera junction boxes and adopted a similar outer casing for our device. The internal layout was carefully designed to secure all electronic components firmly in place. To absorb shocks, we added rubber bushes when mounting the electronics. Additionally, we used two adjustable straps to securely attach the device to the fence pole.

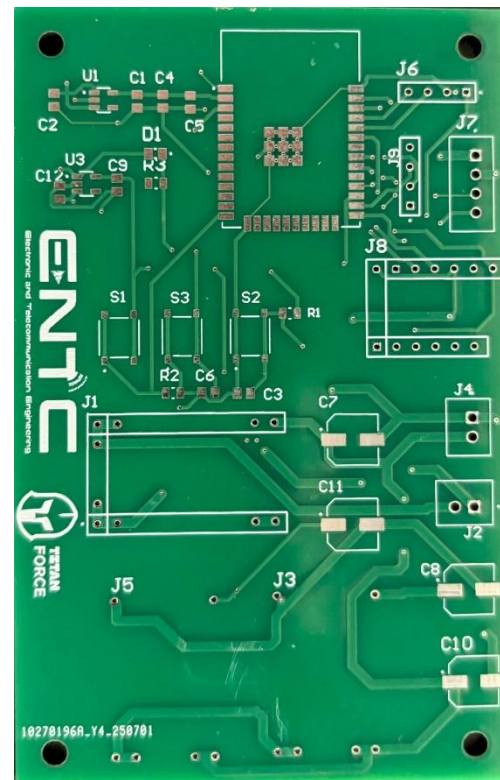
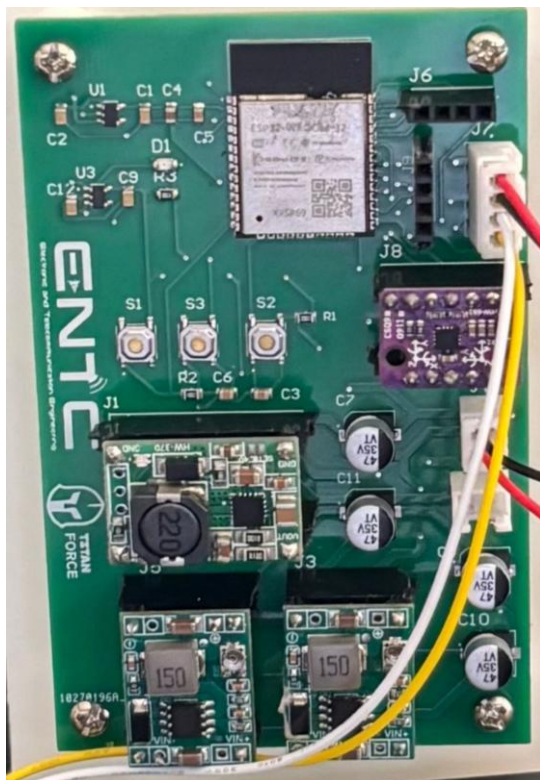
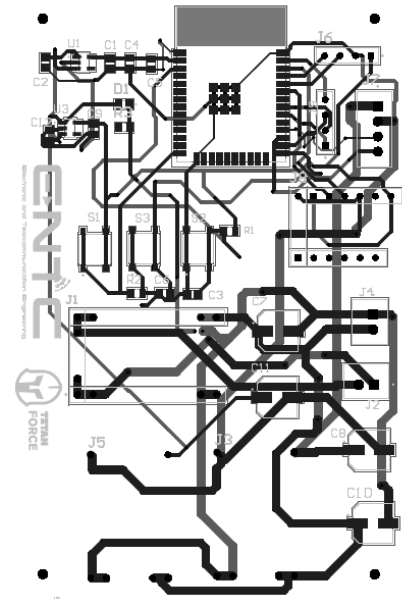
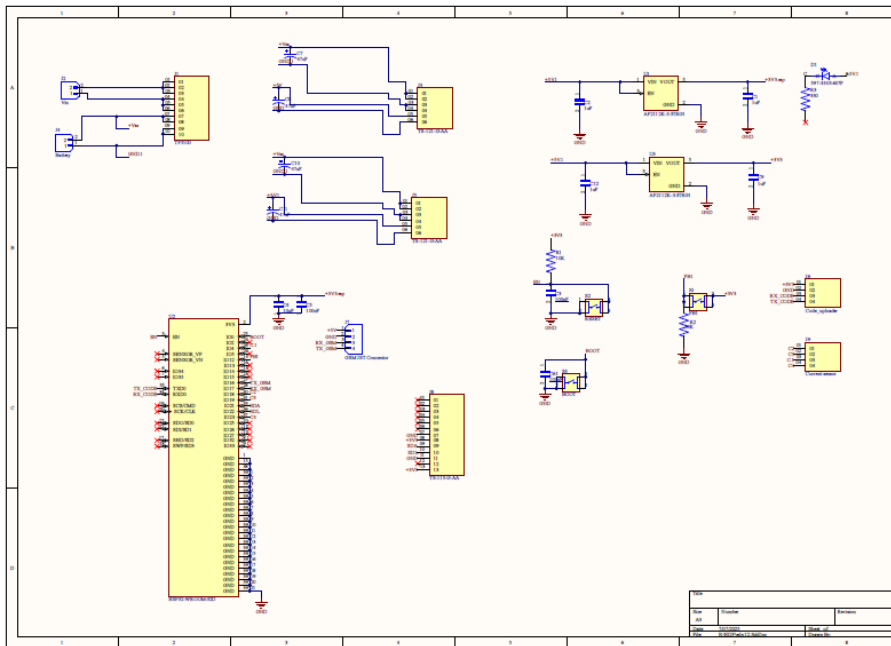
6.1 Final Enclosure Design



6.2 Final Product



7. PCB Design



8. Final Budget

Component	Unit Price (Rs.)	Quantity	Price (Rs)
SMD Components	780		780
ESP 32 Chip	900	1	900
MP1584 Voltage Regulators	300	2	600
TP 5100 battery charging module	300	1	300
3.4V 1800mAh rechargeable battery	380	2	760
5V, 80mA Solar Panels	300	2	
The BMI160 accelerometer gyroscope	350	1	350
GSM 900A Module	1200	1	1200
Enclosure	3700	1	3700
Total			8590

9. Market Analysis

- Government

The government consistently strives to protect both human and elephant lives by seeking solutions to the human-elephant conflict. In addition to the tragic loss of lives, damage to farms and property also has an indirect impact on Sri Lanka's economy. Currently, watch huts are maintained in many areas, where officers (Grama Arakshaka) monitor for elephant breaches—an effort funded by the government. By installing this device, the government can potentially reduce or eliminate these operational costs while improving response efficiency.

- NGO's and Private Companies

During our research, we discovered that several private companies are involved in installing elephant fences. However, most of them lack a real-time monitoring system. The only existing method they use is detecting power failures, which only identifies issues at the power source, not breaches in the fence. We see an opportunity to pitch our

device to these companies to enhance their solutions. In addition, non-governmental organizations (NGOs) that engage in conservation or rural development initiatives may also find this device valuable for their charitable projects.

- Large-scale farmers

Farmers who build and maintain their own elephant fences are also a key market segment for our device. These individuals often suffer the most from property and crop damage and would benefit greatly from a reliable, real-time alert system to protect their livelihoods.

We plan to introduce a single device to the market at a price of LKR 9,000–10,000. However, for bulk purchases, we are able to offer a reduced price.

10. Task Allocation

Name	Task
Karunanayake A.H. D.	Enclosure Design, GSM module programming
Piyumal N. P. P.	PCB design, Circuit design, Gyro module programming
Jayasinghe J. A. P. R.	Mobile app design, Soldering
Withanage G. D. N.	Mobile app backend implementation, Assembly